

SPRINT 4 BACKLOG

ShadowStack: Predictive Cloud Cost Optimizer

FastAPI — PostgreSQL — React + Vite — D3.js — PyTorch — Docker
Product Backlog v1.2 · Sprint 4 of 4 · Final Sprint

Project Team: Aniruddha Bhide (Lead/Backend/ML), Sanay Krishna (Frontend/Docs)

Course: BCA 6th Semester Project

Planned SP (original PBIs): 30

Total Sprint 4 SP: 40

Sprint Duration: 06 Apr – 15 Apr 2026

New Scope (PBI-061–063): +10

Final Submission Deadline: 15 April 2026

Sprint 4 Goal — Final Delivery

Deliver a fully integrated, tested, and documented system ready for academic evaluation. The end-to-end workflow — user logs in via GitHub OAuth → connects a repository → opens a PR → system analyses code and posts cost prediction → dashboard updates in real time — must be live and demonstrable within 30 seconds. All documentation must be 100% complete. All six required demo scenarios must be rehearsed and ready.

New Scope — Identified at End of Sprint 3

Three new backlog items (PBI-061 to PBI-063) have been added based on gaps identified during Sprint 3. These were not in the original backlog but are **mandatory for the end-to-end demo** and have been formally added as new scope. They are marked **NEW** in all tables below.

PBI-061: GitHub OAuth Login (React + Vite) — blocks PBI-062 and the entire auth flow

PBI-062: Settings Page / Repo Connect UI — blocks webhook registration from the frontend

PBI-063: Wire Dashboard to Real Predictions Data — blocks Demo Scenarios 3, 4, and 5

Sprint 4 At a Glance

Sprint	Workstream	Planned SP	Stories	Lead	Deadline
S4	New Scope: OAuth, Settings, Data Wiring (PBI-061–063)	10	3	Both	10 Apr 2026
S4	Integration & E2E Workflow (PBI-047–051)	14	5	Aniruddha	12 Apr 2026
S4	QA, Docs & Final Report (PBI-052–055)	9	4	Aniruddha	14 Apr 2026
S4	UI Polish, Demo Prep & Cleanup (PBI-056–060)	7	5	Both	15 Apr 2026
	Total	40	17		15 Apr

New Scope — OAuth, Settings & Data Wiring

Both Members — Critical Path Blockers

10 SP

ID	Title	User Story	Acceptance Criteria	Pri.	SP
PBI-061 NEW	GitHub OAuth Login	As a user, I want to log in to ShadowStack via my GitHub account so that my repositories and tokens are securely managed.	/login page with GitHub OAuth button; redirects to GitHub with repo + admin:repo_hook scopes; POST /api/auth/github exchanges code for token; token stored; /callback page handles redirect; protected routes redirect unauthenticated users to /login	HIGH	5
PBI-062 NEW	Settings Page / Repo Connect UI	As an authenticated user, I want a settings page where I can connect my GitHub repository so that ShadowStack can analyse my PRs.	/settings page with owner/repo text field and Connect button; button calls POST /api/integrations, with repo + token; success/error feedback shown; connected repos listed	HIGH	3
PBI-063 NEW	Wire Dashboard to Real Predictions Data	As a user, I want the dashboard charts to show real PR analysis results from my connected repository so that I can see actual cost predictions.	useDashboardData.1 passes connected repo as filter; GET /api/predictions? returns scoped data; grouped bar chart shows one bar per PR analysis; mockAdapter only activates when API is truly unreachable	HIGH	2

Integration & End-to-End Workflow

ID	Title	User Story	Acceptance Criteria	Pri.	SP
PBI-047	GitHub PR Comment Automation	As a developer, I want the system to auto-post a cost-prediction comment on each PR so that I get feedback without leaving GitHub.	On PR open/update: markdown cost table (baseline, predicted, delta, recommendations) posted to correct PR within 30 s; verified on live test repository	HIGH	5
PBI-048	End-to-End PR to Dashboard Workflow	As a developer, I want to open a PR and see results on the dashboard within 30 seconds so that the full pipeline is operational.	PR opened → webhook fires → HMAC validates → BackgroundTask runs → prediction stored → PR comment posted → dashboard polls and updates; all within 30 s; documented with screenshots	HIGH	5
PBI-049	Performance Optimisation & Profiling	As a developer, I want critical paths profiled and bottlenecks resolved so that all SLO time targets are met.	PR analysis <30 s; API P95 <500 ms; dashboard load <2 s; profiling report committed to repo	HIGH	2
PBI-050	Final Docker Deployment Verification	As a developer, I want to confirm the full system starts from a clean clone with one command so that deployment is reproducible.	git clone + docker compose up = working system on a fresh machine; verified by Sanay independently	HIGH	1
PBI-051	API Documentation	As an evaluator, I want complete API documentation so that all endpoints are clearly understood.	All endpoints in OpenAPI at /docs; Postman collection exported; example requests and responses included	HIGH	1

QA, Documentation & Final Report

Aniruddha Bhide — QA & Documentation

9 SP

ID	Title	User Story	Acceptance Criteria	Pri.	SP
PBI-052	Integration Tests: Critical Workflows	As a QA engineer, I want automated integration tests covering the webhook-to-prediction path so that critical workflows are verified.	Covers: webhook → analysis → DB write → API response; all pass in CI; results documented in repo	HIGH	3
PBI-053	System Testing & Validation	As a QA engineer, I want the system validated against all measurable criteria so that targets are demonstrably met.	Verified: ML MAE <15%, PR response <30 s, load <2 s, uptime ≥95%, coverage ≥70%; validation report submitted	HIGH	3
PBI-054	Deployment Guide & User Manual	As a user, I want step-by-step setup and usage documentation so that evaluators can run ShadowStack independently.	Docker setup, env vars, DB migration, test run; dashboard and GitHub integration usage guide included	HIGH	1
PBI-055	Final Project Report Compilation	As a team, I want a consolidated final report combining all documentation and results so that the project is ready for submission.	Includes: executive summary, architecture, ML results, test results, limitations, future scope; reviewed by both members	HIGH	2

UI Polish, Demo Preparation & Code Cleanup

Sanay Krishna — UI Polish & Demo

ID	Title	User Story	Acceptance Criteria	Pri.	SP
PBI-057	UI/UX Refinements & Final Polish	As a user, I want the dashboard to feel polished and professional so that it makes a strong impression at evaluation.	Consistent colour scheme; no layout overflow; loading/error states on all async ops; cross-browser verified (Chrome, Firefox)	MED	1
PBI-060	Sprint 4 Retrospective & Lessons Learned	As a team, I want a combined lessons-learned document from all four sprints so that Agile artefacts are complete.	Sprint 4 retrospective held; all-sprints document committed; included in final report appendix	MED	1

Both Members — Shared Ownership

ID	Title	User Story	Acceptance Criteria	Pri.	SP
PBI-056	Comprehensive Bug Fixing	As a team, I want all known bugs resolved before submission so that the system is stable during evaluation.	Zero open P1/P2 bugs; all issues closed; regression test added for each fix	HIGH	2
PBI-058	Demo Preparation & Slides	As a team, I want a rehearsed demo and slide deck so that we can present all 6 scenarios confidently.	All 6 demo scenarios rehearsed ≥ 2 times; slide deck complete; backup screenshots prepared	HIGH	2
PBI-059	Code Cleanup & Refactoring	As a developer, I want dead code removed and naming standardised so that the codebase is clean for submission.	No commented-out code; consistent naming (snake_case Python, camelCase JS); no unresolved TODOs; linter passes	MED	1

Sprint 4 Success Criteria

#	Criterion	Related PBIs
1	User can log in via GitHub OAuth and token is securely stored	PBI-061
2	User can connect a repository; webhook auto-registered on GitHub	PBI-062, PBI-047
3	Opening a PR triggers analysis, stores prediction, posts comment	PBI-047, PBI-048
4	Dashboard updates within 30 s of a PR being opened; shows real data	PBI-048, PBI-063
5	End-to-end pipeline verified: login → connect → PR → dashboard update	PBI-048
6	All unit and integration tests pass; coverage $\geq 70\%$	PBI-052, PBI-053
7	PR analysis response time < 30 s; API P95 < 500 ms; load < 2 s	PBI-049
8	System starts from clean clone with <code>docker compose up</code>	PBI-050
9	All 6 demo scenarios rehearsed and ready for live demonstration	PBI-058
10	Final report and all documentation 100% complete	PBI-055, PBI-054

Live Demo — Required Scenarios

#	Scenario	Blocked Until
1	User logs in via GitHub OAuth; session token visible in browser	PBI-061
2	User connects <code>owner/repo</code> ; webhook auto-registered on that repo	PBI-062
3	Developer opens a PR with Python code changes on the connected repo	PBI-047, PBI-048
4	System analyses code and posts a markdown cost-prediction comment on the PR	PBI-047
5	Dashboard updates within 30 s showing real prediction data	PBI-063, PBI-048
6	ML model performance metrics panel showing Accuracy, R^2 , MAPE, MAE, RMSE	Already built (S3)

Definition of Done (Sprint 4)

A Sprint 4 item is **Done** when:

1. All acceptance criteria verified and signed off by item owner.
2. Code merged to `main` via a reviewed GitHub pull request.
3. Relevant unit/integration tests written and passing in CI.
4. No new ESLint (frontend) or Pylint (backend) errors introduced.
5. Feature demonstrated in the running Docker Compose environment.
6. Impacted documentation updated (README, API docs, or user guide).
7. Item marked Done on the Agile project board.

ShadowStack — Sprint 4 Backlog · Product Backlog v1.2 · Apr 6 – Apr 15, 2026
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Final Sprint. Final Deadline: 15 April 2026.